WORKED SOLUTION

STEP III 1988, Question 3

Pure • Complex Loci

STEP III 1988, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s3-q3

The key idea

The map $w = (z+\mathrm{i})/(z-\mathrm{i})$ can be turned round: solving for z gives $z = \mathrm{i}(w+1)/(w-1)$, so any condition on z becomes a condition on w. For the circle, the parametrisation does the work instead, because $z-\mathrm{i}$ is then simply $2\mathrm{e}^{\mathrm{i}\theta}$. For the real and imaginary cases, think of moduli and arguments: a real z is equidistant from i and $-\mathrm{i}$.

Solution

The parametric form

The curve $|z-\mathrm{i}| = 2$ is the circle with centre i and radius 2, so

$$z = \mathrm{i} + 2\mathrm{e}^{\mathrm{i}\theta}, \qquad -\pi < \theta \leq \pi,$$

or, with $z = x + \mathrm{i}y$, $x = 2\cos\theta$ and $y = 1 + 2\sin\theta$.

We'll also use the inverse map. From $w(z-\mathrm{i}) = z+\mathrm{i}$ we get $z(w-1) = \mathrm{i}(w+1)$, so for $w \neq 1$

$$z = \frac{\mathrm{i}(w+1)}{w-1}.$$

No z gives $w = 1$, since that would need $z+\mathrm{i} = z-\mathrm{i}$. Every $w \neq 1$ comes from exactly one $z \neq \mathrm{i}$.

Part (i)

Substituting the parametric form, $z+\mathrm{i} = 2\mathrm{i} + 2\mathrm{e}^{\mathrm{i}\theta}$ and $z-\mathrm{i} = 2\mathrm{e}^{\mathrm{i}\theta}$, so

$$w = \frac{2\mathrm{i} + 2\mathrm{e}^{\mathrm{i}\theta}}{2\mathrm{e}^{\mathrm{i}\theta}} = 1 + \mathrm{i}\,\mathrm{e}^{-\mathrm{i}\theta} = 1 + \mathrm{e}^{\mathrm{i}(\pi/2-\theta)}.$$

As θ runs once round, $\pi/2 - \theta$ also takes every angle once. So w traces the *whole* circle

$$|w-1| = 1,$$

with centre 1 and radius 1. For example, $z = 3\mathrm{i}$ gives $w = 2$, and $z = -\mathrm{i}$ gives $w = 0$. The point $w = 1$ isn't on this circle, which is consistent with the fact that it's never attained.

Part (ii)

If $z = x$ is real, it's the same distance from i as from $-\mathrm{i}$, so $|x + \mathrm{i}| = |x - \mathrm{i}|$ and

$$|w| = \frac{|x + \mathrm{i}|}{|x - \mathrm{i}|} = 1.$$

Explicitly, multiplying top and bottom by $x + \mathrm{i}$,

$$w = \frac{(x + \mathrm{i})^2}{x^2 + 1} = \frac{x^2 - 1}{x^2 + 1} + \mathrm{i}\,\frac{2x}{x^2 + 1}.$$

Ask yourself Is every point of the unit circle reached?

No, not quite. Take any w with $|w| = 1$ and $w \neq 1$, and write $w = \mathrm{e}^{\mathrm{i}\phi}$. Then

$$z = \mathrm{i}\,\frac{\mathrm{e}^{\mathrm{i}\phi} + 1}{\mathrm{e}^{\mathrm{i}\phi} - 1} = \mathrm{i}\,\frac{2\cos(\phi/2)}{2\mathrm{i}\sin(\phi/2)} = \cot\tfrac{\phi}{2},$$

which is real. So every such point is attained, but $w = 1$ never is: it's the limit as $x \to \pm\infty$. The locus is the unit circle $|w| = 1$ with the single point $w = 1$ removed. For instance, $x = 0$ gives $w = -1$, and $x = 1$ gives $w = \mathrm{i}$.

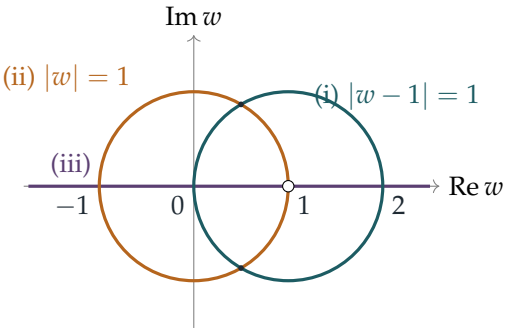
Part (iii)

If $z = \mathrm{i}y$ with y real and $y \neq 1$, then

$$w = \frac{\mathrm{i}(y + 1)}{\mathrm{i}(y - 1)} = \frac{y + 1}{y - 1} = 1 + \frac{2}{y - 1},$$

which is real. As y ranges over the reals other than 1, the fraction $2/(y - 1)$ takes every non-zero real value, so w takes every real value except 1. (Conversely, if $w = u$ is real with $u \neq 1$, then $z = \mathrm{i}(u + 1)/(u - 1)$ is imaginary.) The locus is the real axis with the point $w = 1$ removed. Here $z = 0$ gives $w = -1$; if you read "imaginary" as excluding $z = 0$, then $w = -1$ is removed as well.

Sketch



The teal circle is the locus for (i), the amber circle (minus the point 1) is the locus for (ii), and the purple axis (minus the point 1) is the locus for (iii). The open dot marks $w = 1$, which is never attained.

Checks

- The circle in (i) meets the real axis at $z = \pm\sqrt{3}$. These should map to points on both the teal and amber circles, and indeed $w(\sqrt{3}) = (\sqrt{3}+\mathrm{i})^2/4 = \frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i} \approx 0.5 + 0.866\mathrm{i}$, which lies on $|w| = 1$ and on $|w - 1| = 1$.
- The circle in (i) meets the imaginary axis at $3\mathrm{i}$ and $-\mathrm{i}$, which map to 2 and 0: exactly where the teal circle crosses the real axis.
- A numerical check of 10^4 random points on each locus confirmed $|w - 1| = 1$, $|w| = 1$ and $\operatorname{Im} w = 0$, and showed that the inverse map recovers z.

Common slips

- Forgetting that $w = 1$ is missing in (ii) and (iii). It corresponds to "$z = \infty$".
- Expanding into x and y in (i) and getting lost in the algebra. The parametrisation makes $z - \mathrm{i}$ a single exponential, and the rest is one line.

Beyond STEP

The map $z \mapsto (z+\mathrm{i})/(z-\mathrm{i})$ is a Möbius transformation. Such maps always send circles and lines to circles and lines, which is exactly what you found in all three parts. They are best understood on the Riemann sphere, where adding a "point at infinity" fills the gap at $w = 1$. Its relative, the Cayley transform $z \mapsto (z-\mathrm{i})/(z+\mathrm{i})$, maps the upper half-plane onto the unit disc. Needham's *Visual Complex Analysis* has a beautiful geometric treatment of all of this.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.